Experiment-7

Write a python program to implement BFS

Aim

To write and execute a Python program to traverse a graph using Breadth-First Search (BFS), which visits the nodes level by level starting from a given node.

Algorithm

1. Represent the graph as an adjacency list.
2. Create an empty queue and insert the start node A into it.
3. Mark the start node as visited.
4. Remove the first node from the queue and add it to the traversal order.
5. Insert all unvisited neighbours of that node into the queue and mark them as visited.
6. Repeat steps 4 and 5 until the queue becomes empty.
7. Display the BFS traversal order.

Coding

ex-7.py - D:/ex-7.py (3.10.0)

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```
# Breadth First Search (BFS)
from collections import deque

graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": ["F"],
    "F": []
}

def bfs(start):
    visited = [start]
    queue = deque([start])
    order = []

    while queue:
        node = queue.popleft()
        order.append(node)

        # Add all unvisited neighbours to the queue
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.append(neighbour)
                queue.append(neighbour)

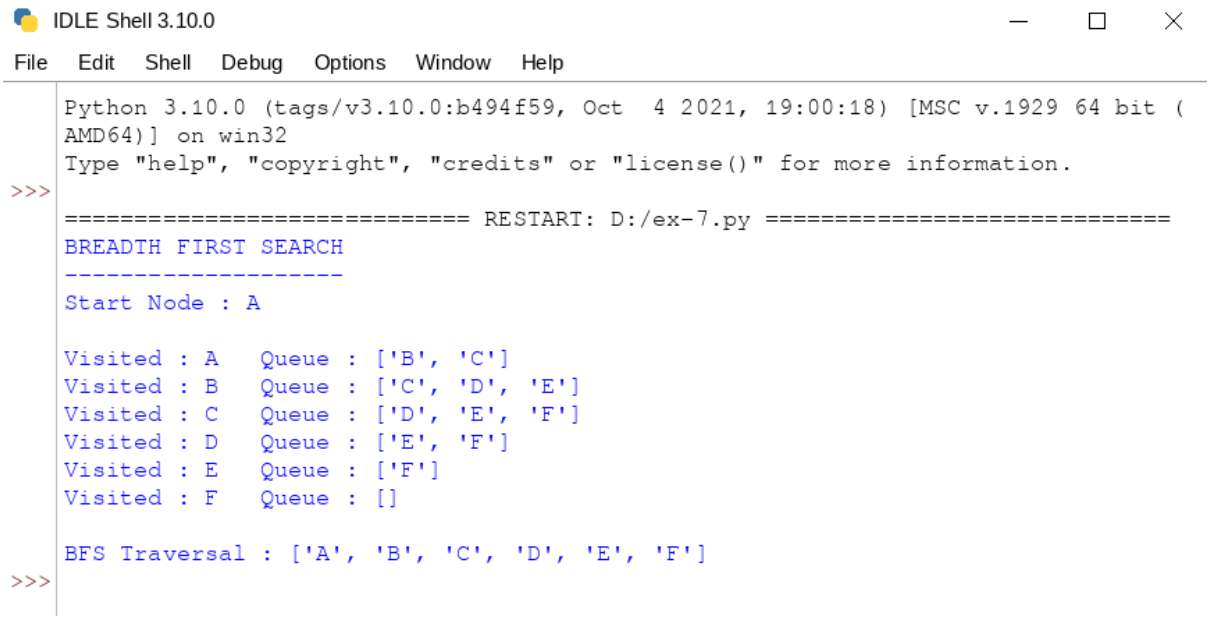
        print("Visited :", node, "  Queue :", list(queue))
    return order
```

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```
print("BREADTH FIRST SEARCH")
print("--------------------")
print("Start Node : A")
print()
result = bfs("A")
print()
print("BFS Traversal :", result)
```

Output



Result

The Breadth-First Search (BFS) algorithm was successfully implemented. The program traversed the graph level by level and produced the order A, B, C, D, E, F.